Yes. The paper is Wang and Samworth, **"High dimensional change point estimation via sparse projection"**, JRSSB 80(1), 57–83. The way Section 3 is written makes the method look considerably more complicated than its central idea actually is. The whole paper can be understood as:

> **Learn one direction in $\mathbb{R}^p$ that combines the coordinates containing the change, project the high-dimensional series onto that direction, and then solve an ordinary one-dimensional change-point problem.**

That is essentially INSPECT. [source]

First: what on earth is $M_{j,r}$?

You are right that this is unnecessarily easy to get stuck on.

In equation (8), they define a **generic CUSUM transformation**

$$\mathcal{T}_{p,n} : \mathbb{R}^{p\times n} \longrightarrow \mathbb{R}^{p\times(n-1)}$$

and write

$$[\mathcal{T}_{p,n}(M)]_{j,t} = \sqrt{\frac{t(n-t)}{n}}\left[\frac{1}{n-t}\sum_{r=t+1}^{n} M_{j,r} - \frac{1}{t}\sum_{r=1}^{t} M_{j,r}\right].$$

Here,

$$\boxed{M_{j,r} = \text{the entry in row } j \text{ and column } r \text{ of the generic matrix } M.}$$

That's it.

Think of a $p \times n$ data matrix

$$X = \begin{pmatrix} X_{1,1} & X_{1,2} & \cdots & X_{1,n} \\ X_{2,1} & X_{2,2} & \cdots & X_{2,n} \\ \vdots & \vdots & & \vdots \\ X_{p,1} & X_{p,2} & \cdots & X_{p,n} \end{pmatrix}.$$

- $j = 1, \ldots, p$ = **coordinate / component series**
- $r = 1, \ldots, n$ = **time**
- t = **candidate change-point location**

So when they actually use the transformation with $M = X$,

$$M_{j,r} = X_{j,r}.$$

For example, $X_{7,40}$ is the observation of the **7th component at time 40**.

Their M in (8) is just a placeholder so they can later write

$$T = \mathcal{T}(X), \qquad A = \mathcal{T}(\mu), \qquad E = \mathcal{T}(W).$$

The paper explicitly decomposes $X = \mu + W$, and by linearity gets $T = A + E$. [source]

There is one annoying additional issue: **later they reuse the letter M** as an optimization variable in the sparse-SVD convex relaxation. That M is a $p \times (n-1)$ matrix and is conceptually different from the generic input matrix M in equation (8). I would absolutely rename one of them in your class notes.

The paper reconstructed in a sensible order

Suppose

$$X_t \in \mathbb{R}^p, \qquad t = 1, \dots, n,$$

and

$$X_t = \mu_t + \varepsilon_t.$$

Initially they assume

$$\varepsilon_t \sim N_p(0, \sigma^2 I_p).$$

The mean is piecewise constant. For a single change point z,

$$\mu_t = \begin{cases} \mu^{(0)}, & t \le z, \\ \mu^{(1)}, & t > z. \end{cases}$$

Define the change vector

$$\theta = \mu^{(1)} - \mu^{(0)}.$$

The high-dimensional assumption is that only a few coordinates change:

$$\|\theta\|_0 \le k, \qquad k \ll p.$$

So maybe $p = 1000$, but only 20 coordinates undergo a mean shift. [source]

---

1. If we knew θ, the problem would be easy

This is really the starting point of the method.

Take any unit vector $v \in \mathbb{R}^p$ and project:

$$Y_t = v^\top X_t.$$

Before the change its mean is

$$v^\top \mu^{(0)}$$

and afterwards it is

$$v^\top \mu^{(1)}.$$

Therefore the size of the projected change is

$$v^\top \theta.$$

Among all $\|v\|_2 = 1$, this is maximized by

$$\boxed{v_{\text{oracle}} = \frac{\theta}{\|\theta\|_2}.}$$

Then the size of the change becomes

$$v_{\text{oracle}}^\top \theta = \|\theta\|_2.$$

This is why projection helps. If 30 coordinates each have a small shift, you can combine them into **one one-dimensional series with a much stronger signal**. The paper explicitly identifies this normalized change vector as the oracle projection direction. [source]

So the whole problem becomes:

$$\boxed{\text{How do we estimate the direction } \theta/\|\theta\|_2?}$$

---

2. Construct a CUSUM matrix

For every coordinate j and every potential split t, calculate

$$T_{j,t} = \sqrt{\frac{t(n-t)}{n}} \left(\underbrace{\bar{X}_{j,(t+1):n}}_{\text{mean after } t} - \underbrace{\bar{X}_{j,1:t}}_{\text{mean before } t} \right).$$

Thus

$$T \in \mathbb{R}^{p \times (n-1)}.$$

A much clearer way of thinking about this matrix is:

	$t=1$	$t=2$	$\cdots$	$t=n-1$
$j=1$	CUSUM	CUSUM	$\cdots$	CUSUM
$j=2$	CUSUM	CUSUM	$\cdots$	CUSUM
$\vdots$				
$j=p$	CUSUM	CUSUM	$\cdots$	CUSUM

Each **column** t asks:

> "How much evidence is there for a change at time t, coordinate by coordinate?"

Each **row** j asks:

> "What does the CUSUM curve look like for component j?"

This is equation (8). [source]

---

3. The beautiful observation in the paper: population CUSUM is rank one

This is the actual mathematical trick.

Imagine that instead of noisy X, we knew the population mean matrix

$$\mu = (\mu_1, \ldots, \mu_n).$$

Apply exactly the same CUSUM transformation:

$$A = \mathcal{T}(\mu).$$

For a single change point, Wang and Samworth show that

$$\boxed{A = \theta\gamma^\top}$$

for a certain $(n-1)$-vector γ determined by the change-point location z. [source]

This equation explains almost the entire paper.

Because

$$A = \theta\gamma^\top$$

is an outer product, A has **rank one**.

Its left singular direction is

$$\frac{\theta}{\|\theta\|_2}.$$

So:

oracle projection direction = leading left singular vector of the population CUSUM matrix.

That is the key insight.

Why does $A = \theta\gamma^\top$ make sense?

Suppose coordinate j changes by θ_j.

Its population CUSUM curve has some universal time shape:

$$\gamma_1, \gamma_2, \ldots, \gamma_{n-1},$$

peaking around the true change point z.

Coordinate j's curve is simply that shape multiplied by θ_j:

$$A_{j,t} = \theta_j \gamma_t.$$

Thus

$$\begin{gathered}\text{CUSUM row 1} = \theta_1\gamma^\top \\ \text{CUSUM row 2} = \theta_2\gamma^\top \\ \vdots \\ \text{CUSUM row p} = \theta_p\gamma^\top.\end{gathered}$$

All rows are proportional.

Hence rank one.

I would make **this** the centerpiece of your class explanation rather than starting with their optimization problem.

---

4. But we observe $T = A + E$, not A

From

$$X = \mu + W$$

we get

$$T = \mathcal{T}(X) = \mathcal{T}(\mu) + \mathcal{T}(W) = A + E.$$

So we would like to recover the leading left singular vector of the signal matrix A from noisy T.

A naive solution is:

$$\hat{v} = \text{leading left singular vector of } T.$$

But when p is enormous, lots of irrelevant coordinates contribute noise.

The authors know something useful:

$$\theta \text{ is sparse.}$$

Therefore the desired singular vector should itself be sparse.

---

5. Ideally: sparse SVD

Conceptually, they want

$$\hat{v} = \arg \max_{\substack{\|v\|_2=1 \\ \|v\|_0 \le k}} \|T^\top v\|_2.$$

Interpretation:

> Find a unit vector involving only a few coordinates such that the resulting linear combination of their CUSUM curves has as much energy as possible.

This is precisely a **sparse leading left singular vector** problem. [source]

Unfortunately, solving that directly is combinatorial/NP-hard.

That's why the paper suddenly wanders into convex optimization.

---

6. The convex optimization is just a computational surrogate

They introduce a matrix optimization problem of roughly the form

$$\boxed{\hat{M} = \arg \max_{M \in \mathcal{S}} \{\langle T, M \rangle - \lambda \|M\|_1\}.}$$

The two pieces have simple interpretations.

Fit the CUSUM matrix

$$\langle T, M \rangle$$

rewards M for aligning with the observed CUSUM signal.

Impose sparsity

$$-\lambda\|M\|_1$$

discourages lots of nonzero entries and therefore suppresses noisy coordinates.

The nuclear-norm constraint

$$\|M\|_* \le 1$$

is their convex replacement for the desired rank-one constraint.

So don't present this as some mysterious standalone optimization problem. It is simply:

$$\boxed{\text{sparse rank-one approximation of the CUSUM matrix.}}$$

The direct sparse-SVD problem is hard, so they replace the rank and ℓ_0 constraints by convex surrogates: a nuclear-norm constraint and an ℓ_1 penalty. [source]

They use ADMM to solve one version. Algorithm 1 is essentially **a numerical solver**, not a new statistical idea.

---

7. Extract the projection direction

Once they obtain $\hat{M}$, calculate its leading left singular vector:

$$\boxed{\hat{v} = \text{leading left singular vector of } \hat{M}.}$$

This estimates

$$v_{\text{oracle}} = \frac{\theta}{\|\theta\|_2}.$$

This is Steps 2–3 of their Algorithm 2. [source]

---

8. Now throw away the high dimension

Once $\hat{v}$ has been found, the hard high-dimensional part is finished.

Project:

$$Y_t = \hat{v}^\top X_t.$$

Equivalently, because the CUSUM transform is linear,

$$\text{CUSUM}(Y)_t = \hat{v}^\top T_t,$$

where T_t is column t of the CUSUM matrix.

Then simply choose

$$\boxed{\hat{z} = \arg\max_{1 \le t \le n-1} |\hat{v}^\top T_t|.}$$

That is an entirely ordinary one-dimensional CUSUM estimator.

The associated test statistic is

$$\bar{T}_{\max} = |\hat{v}^\top T_{\hat{z}}|.$$

This is Step 4 of Algorithm 2. [source]

So Algorithm 2 can be rewritten much more cleanly

Given

$$X = (X_1, \dots, X_n) \in \mathbb{R}^{p \times n},$$

do:

1. **CUSUM all coordinates**
$$T_{j,t} = \sqrt{\frac{t(n-t)}{n}} (\bar{X}_{j,t+1:n} - \bar{X}_{j,1:t}).$$
2. **Find the sparse direction in which these CUSUM curves line up**
$$T \quad \longrightarrow \quad \hat{v}.$$
Technically this is done by their convex sparse-SVD relaxation.
3. **Project**
$$Y_t = \hat{v}^\top X_t.$$
4. **Run an ordinary univariate CUSUM**
$$\hat{z} = \arg\max_t |\,\mathrm{CUSUM}(Y)_t\,|.$$

That's INSPECT for one change point.

What Algorithm 3 is about — and why I would barely mention it in class

They then introduce the odd/even sample split:

$$X^{(1)}_{j,t} = X_{j,2t-1}, \qquad X^{(2)}_{j,t} = X_{j,2t}.$$

Use one half to estimate $\hat{v}$, and the other half to estimate z.

Why?

Not because it improves the method.

It makes the **proof easier**, since the estimated direction and the observations subsequently used to localize the change become independent.

The authors explicitly say that in practice they recommend Algorithm 2 using all the data rather than sample splitting. [source]

So for a class talk:

> "They analyze a sample-split variant for technical reasons, but the practical INSPECT algorithm does not split the sample."

Done.

Multiple change points: INSPECT + wild binary segmentation

The single-change theory assumes an interval contains one relevant change.

If the full series has many changes, running it once on the whole interval can fail because different changes can cancel or involve different coordinates.

So they use **wild binary segmentation (WBS)**.

Generate many random intervals

$$(s_q, e_q), \qquad q = 1, \ldots, Q.$$

For each interval,

$$X_{s_q+1:e_q},$$

run the single-change INSPECT procedure.

Each interval gives:

$$(\hat{z}^{[q]}, \bar{T}^{[q]}_{\max}).$$

Choose the interval with the largest statistic:

$$q_0 = \arg\max_q \bar{T}^{[q]}_{\max}.$$

The candidate change in the original time scale is

$$b = s_{q_0} + \hat{z}^{[q_0]}.$$

If

$$\bar{T}^{[q_0]}_{\max} > \xi,$$

declare b to be a change point and recurse on

$$[s, b] \quad \text{and} \quad [b, e].$$

If it is below ξ, stop. [source]

The reason WBS is especially useful here is subtle and important: two different high-dimensional changes may occur in **different coordinate subsets**. A projection estimated from a long interval containing both may point toward some compromise that matches neither well. Random intervals have a good chance of isolating individual changes. The authors make exactly this point. [source]

The entire paper in one diagram

$$\boxed{X_t \in \mathbb{R}^p}$$

$$\downarrow$$

coordinatewise CUSUM

$$\boxed{T \in \mathbb{R}^{p\times(n-1)}}$$

$$\downarrow$$

sparse low-rank structure

$$\boxed{\hat{v} \approx \frac{\theta}{\|\theta\|_2}}$$

$$\downarrow$$

project

$$\boxed{Y_t = \hat{v}^\top X_t \in \mathbb{R}}$$

$$\downarrow$$

ordinary univariate CUSUM

$$\boxed{\hat{z} = \arg\max_t |\,\mathrm{CUSUM}(Y)_t\,|}$$

and for many changes, wrap the whole thing inside WBS.

What I would actually say in your PhD class

You could summarize the contribution in about 45 seconds as:

> Wang and Samworth consider high-dimensional mean change points where only a sparse subset of coordinates may change. Their main insight is that, for a single change, the population matrix of coordinatewise CUSUM statistics has rank one: it can be written as $A = \theta\gamma^\top$, where θ is the vector of mean changes. Consequently, the leading left singular vector of this matrix is precisely the optimal projection direction $\theta/\|\theta\|_2$. Since we only observe a noisy CUSUM matrix, they estimate this direction using a convex relaxation of sparse singular-vector estimation. They then project the p-dimensional time series onto the estimated direction and locate the change using an ordinary univariate CUSUM. Multiple change points are handled by embedding this single-change procedure within wild binary segmentation.

That, in my view, is a much better summary of the paper than walking through Algorithms 1–4 in their order.

One final notation warning for your notes: I would rename the generic matrix in equation (8) to B, so that

$$[\mathcal{T}(B)]_{j,t} = \sqrt{\frac{t(n-t)}{n}}\left(\frac{1}{n-t}\sum_{r=t+1}^{n} B_{j,r} - \frac{1}{t}\sum_{r=1}^{t} B_{j,r}\right),$$

and reserve M for their later optimization variable. **That one change removes a surprisingly large amount of the notational confusion.**